

Authors: Felix Omondi & Ezekiel Ekanem

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GitHub Link: <https://github.com/Felomondi/AI-and-future-of-work>

AI, Gender, and the Future of Work: Predicting Occupational Exposure to Artificial Intelligence

1. Introduction & Motivation

The rapid advancement of artificial intelligence is fundamentally restructuring the global labor market, yet the resulting shifts in employment are unlikely to be distributed equitably. The seminal work of Frey and Osborne (2017) first estimated that 47% of U.S. employment was highly susceptible to computerization. While early automation largely impacted routine manual and cognitive tasks, the emergence of Large Language Models (LLMs) and generative AI has shifted the focus toward high-skill, white-collar occupations previously considered automation-proof.

Recent assessments by the International Labour Organization (2025) suggest a pronounced gender dimension to this shift, finding that women in high-income countries are nearly three times more likely than men to hold positions at the highest risk of AI-driven automation. This disparity is often attributed to the overrepresentation of women in clerical, administrative, and service roles that align with current AI capabilities. Furthermore, Felten, Raj, and Seamans (2021) introduced the AI Occupational Exposure (AIOE) index, which bridges the gap between AI technical capabilities and O*NET occupational tasks. Their framework allows for a granular analysis of how specific breakthroughs, such as natural language processing or image recognition, map onto the daily responsibilities of various professions.

While the quantification of AI exposure at the occupational level is a significant step forward, critical gaps remain in understanding the why behind these exposure scores. Much of the existing literature focuses on aggregate employment effects, paying less attention to the specific latent job characteristics, such as social perceptiveness, fine motor skills, or deductive reasoning, that act as either shields or magnets for AI exposure. Moreover, there is an urgent need to examine how this exposure intersects with racial demographics. Research by Muro et al. (2019) at the Brookings Institution indicates that young workers and Hispanics are often overrepresented in occupations with higher automation potential, yet few studies have applied the more nuanced AIOE index to these specific demographic segments.

This project addresses these gaps by using machine learning to achieve two primary objectives: first, to identify which specific occupational traits and skill requirements are the strongest predictors of AI exposure; and second, to determine if the burden of this exposure falls disproportionately on women and racial minorities. If occupations dominated by marginalized groups are fundamentally more exposed to AI, the AI revolution threatens to widen existing wage gaps and exacerbate structural wealth inequality.

By isolating these predictors, our analysis provides empirical evidence for ongoing policy debates regarding targeted workforce retraining, educational reform, and the societal cost-benefit analysis of rapid AI integration.

Research Questions

- **RQ1:** Can we predict an occupation's level of AI automation exposure based on its task characteristics, skill requirements, and work context features?
- **RQ2:** Does the demographic composition of an occupation (gender and race) significantly predict AI exposure after controlling for job tasks and skills?

Expectations

We expect occupations requiring routine cognitive tasks (data entry, information processing) to show higher AI exposure, while those requiring physical manipulation, social-emotional skills, or creative judgment will show lower exposure. For RQ2, we hypothesize that female-dominated occupations may show higher exposure to language-modeling AI, given women's overrepresentation in clerical and administrative roles. If this pattern holds after controlling for occupation characteristics, it suggests the AI transition could widen existing labor market inequalities. Null findings would also be informative: if demographic composition is unrelated to AI exposure after controlling for job content, equity interventions should focus on skill development rather than demographic targeting.

2. Data Description

Data Sources & Collection

Our research design relies on the integration of four primary datasets, each providing a different dimension of the occupational landscape in the United States. The unit of analysis for this study is the **6-digit Standard Occupational Classification (SOC) code**, representing distinct professional roles.

1. **O*NET Database (Version 30.2):** Collected by the U.S. Department of Labor, this is the primary source for predictor variables. We extracted approximately 640 features encompassing Skills (e.g., Active Listening), Abilities (e.g., Oral Expression), Work Activities (e.g., Analyzing Data), and Work Context (e.g., Indoors, Environmentally Controlled). These metrics represent importance or level scores, typically on a 0–7 or 1–5 scale.
2. **Felten et al. AI Occupational Exposure (AIOE) Index:** Developed by researchers at Princeton, NYU, and UPenn (2021), these scores serve as our target variables. The dataset provides three specific metrics: General AIOE, Language Modeling AIOE (LMOE), and Image Generation AIOE (IGOE). These scores were generated by mapping 10 AI applications (like translation or image recognition) to 52 O*NET abilities.
3. **BLS Current Population Survey (CPS, 2025):** This provides the demographic shares for each occupation, including pct_women, pct_black, pct_hispanic, and pct_asian. This data is critical for our second research question regarding the distributional impacts of AI.
4. **BLS Occupational Employment and Wage Statistics (OEWS, 2024):** This dataset provides economic context, specifically Median Annual Wages and Total Employment counts, allowing us to control for the economic status of an occupation.

Data Preparation & Cleaning

The cleaning pipeline, documented in `data_cleaning.ipynb`, followed a tidy data philosophy to ensure each row represents a single occupation and each column a single attribute.

- **Standardization of SOC Codes:** ONET data is often reported at an 8-digit level (e.g., 11-1011.00), while BLS and AIOE data use 6-digit codes (e.g., 11-1011). We truncated all ONET codes and aggregated the data to the 6-digit level to ensure a successful relational merge.
- **Reshaping and Merging:** We pivoted multiple O*NET files from a "long" format (multiple rows per occupation) to a "wide" format (one row per occupation with hundreds of columns) before performing an inner join across all four sources.
- **Feature Engineering & Filtering:** We instituted a strict 20% threshold for missing values; any O*NET feature missing more than 20% of its data was dropped. Crucially, we implemented a list for target variables (AIOE, LMOE, IGOE) and demographic predictors (`pct_women`, etc.) to ensure these were not accidentally dropped during the filtering phase.
- **Imputation & Scaling:** Any remaining null values in the 631 occupations were filled using median imputation to maintain the distribution of the features without being skewed by outliers. Finally, we applied a `StandardScaler` to the O*NET predictors to transform them to $\text{mean}=0$ and $\text{variance}=1$, which is a mathematical requirement for the Lasso and Ridge regularized models used in our analysis.

Variables for Modeling

The final dataset consists of 650 variables. Our modeling strategy utilizes these as follows:

- **Target Variables:** We primarily model AIOE, but also run parallel analyses on LMOE and IGOE to determine if different job skills predict exposure to different types of AI (e.g., do verbal skills predict LMOE while spatial skills predict IGOE?).
- **Primary Predictors:** The 640+ O*NET task/skill features. These are included to answer RQ1, identifying which specific job functions drive AI exposure.
- **Demographic Controls:** The BLS demographic shares (`pct_women`, `pct_black`, etc.) are used for RQ2. We include these to see if they retain predictive power even after controlling for the technical tasks of the job, which would suggest that occupational segregation plays a role in who is exposed to the AI transition.
- **Economic Controls:** Median wages and employment levels are included to differentiate between high-exposure high-wage roles (like software engineers) and high-exposure low-wage roles (like administrative clerks).

3. Methods

To address our research questions, we employed a multi-stage modeling framework that combines predictive performance with robust interpretability and causal inference. Our approach transitioned from unregularized linear baselines to regularized regression, advanced tree-based ensemble methods, and manifold learning. All models were evaluated using a 5-fold cross-validation strategy paired with randomized search for hyperparameter optimization to ensure generalizability.

3.1 Predictive Modeling and Regularization (RQ1)

Our initial step toward predicting AI Occupational Exposure (AIOE) utilized an unregularized Ordinary Least Squares (OLS) regression baseline. While this initially yielded a near-perfect R^2 of 0.98, we quickly diagnosed this as data leakage stemming from the inclusion of alternative AI exposure indices (LMOE and IGOE) in the feature set. After strictly isolating our predictors, the corrected OLS baseline achieved a test R^2 of just 0.38. This sharp drop justified our primary methodological challenge: the O*NET dataset presents a high-dimensional feature space (approximately 640 variables) characterized by extreme multicollinearity, which severely limits the efficacy of standard linear models.

To handle this inherent multicollinearity and high dimensionality, we implemented **Lasso (L1) and Ridge (L2) regressions**.

- **Justification:** Lasso facilitates automated feature selection by shrinking non-essential coefficients to exactly zero, isolating the most critical variables. Ridge stabilizes the model by penalizing the square of the coefficients, which is highly effective when dealing with correlated features.
- **Summary Statistics:** Lasso successfully resolved our dimensionality problem, shrinking 567 redundant O*NET features to zero. The resulting model utilized only 71 core predictors and achieved a highly robust test R^2 of 0.975.

3.2 Non-Linear Ensembles (RQ1)

To capture complex feature interactions and non-linear relationships that linear models inherently overlook, we implemented **Random Forest** and **XGBoost** regressors.

- **Justification:** While regularized linear models provide excellent baseline interpretability, human labor tasks often have complex, intersecting boundaries (e.g., the combination of physical stamina and fine motor skills) that tree-based methods are better suited to map.
- **Summary Statistics:** Both models demonstrated exceptional predictive power. Random Forest achieved a cross-validated R^2 of 0.961 utilizing deep decision trees. However, XGBoost emerged as our highest-performing model overall, achieving an R^2 of 0.970 with a remarkably low Mean Absolute Error (MAE) of 0.121. XGBoost achieved this superior accuracy by methodically building shallow, weak trees (maximum depth of 3) and utilizing a slow learning rate to sequentially correct past errors.

3.3 Interpretability and Manifold Learning

Because ensemble models operate as "black boxes," we required mathematically grounded methods to extract insights from our highest-performing XGBoost model.

SHAP (Shapley Additive Explanations): We applied SHAP values to allocate how each specific occupational characteristic shifted an occupation's AIOE score. SHAP provides a unified measure of feature importance that guarantees consistent attribution.

UMAP (Uniform Manifold Approximation and Projection):

To better understand the structural distribution of AI exposure across the labor market, we executed a UMAP projection to compress the 640-dimensional O*NET task space into a two-dimensional embedding.

- **Justification:** UMAP allows us to visualize natural groupings of occupations based solely on task similarity, completely blind to the target variable.

3.4 Causal Inference for Demographic Disparity (RQ2)

To investigate whether demographic composition (e.g., percentage of women or racial minorities) significantly predicts AI exposure after controlling for job tasks, we moved beyond simple correlation to implement **Double Machine Learning (DML)**.

- **Justification:** DML allows us to estimate the causal treatment effect of demographic composition on AI exposure while high-dimensionally controlling for the 640 O*NET job tasks and skills. It does this by using two auxiliary models (one predicting demographic share, the other predicting the AIOE score) and regressing the residuals. This ensures that any observed demographic disparity is not merely a byproduct of confounding variables, such as differing baseline skill distributions.
- **Methodological Refinement:** Based on the success of our XGBoost model for RQ1, we refined our original DML approach. Instead of relying on linear DML, we utilized non-linear, tree-based models as the auxiliary predictors within the DML framework to better isolate the causal impact of demographic composition.
- **Evaluation:** For the DML component, we evaluated our findings by extracting point estimates and 95% confidence intervals to determine the strict statistical significance of demographic disparities in AI exposure.

4. Results

This section presents the findings from our machine learning pipeline, designed to uncover the specific occupational traits that predict AI exposure and to evaluate how this exposure is distributed across different demographic groups.

4.1 Primary Findings: The Drivers of AI Exposure (RQ1)

Our first objective was to determine if we could accurately predict an occupation's AI Occupational Exposure (AIOE) based solely on its task and skill profile. By applying regularized linear models and advanced tree-based ensembles to the 640-dimensional O*NET dataset, we achieved highly accurate predictive performance and extracted a clear narrative regarding automation vulnerability.

Model Performance and Feature Selection

Our highest-performing linear model, Lasso (L1) regression, achieved a remarkable test R^2 of 0.975. By automatically shrinking 567 redundant features to zero, the Lasso model isolated a core group of just 71 highly predictive job traits. However, our optimized XGBoost regressor ultimately proved to be the most robust engine for capturing complex task boundaries, achieving an R^2 of 0.970 with a remarkably low

Mean Absolute Error (MAE) of 0.121. XGBoost accomplished this by utilizing shallow, "weak" trees (maximum depth of 3) and a slow learning rate, which sequentially corrected errors without overfitting.

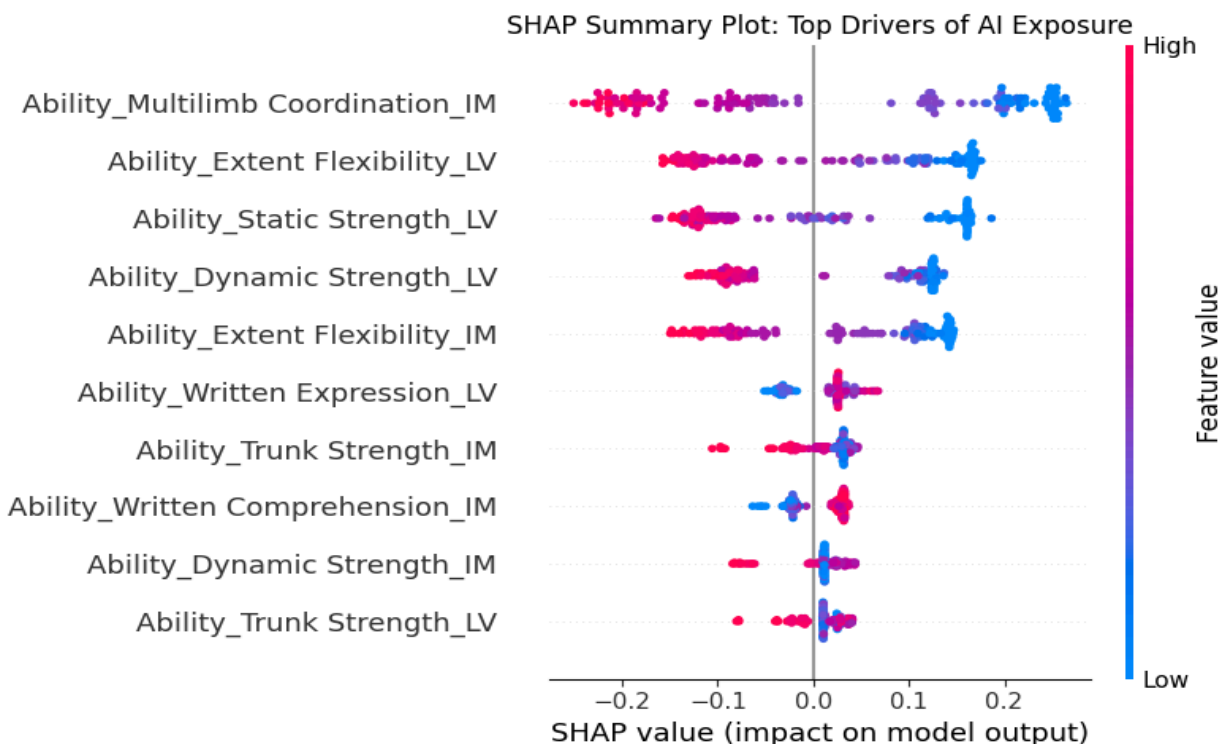
Interpretation of the Feature Space (SHAP)

To answer *what* drives these predictions, we applied SHAP (Shapley Additive Explanations) to our optimized XGBoost model. The SHAP analysis revealed a stark, fundamental divide between cognitive and physical labor:

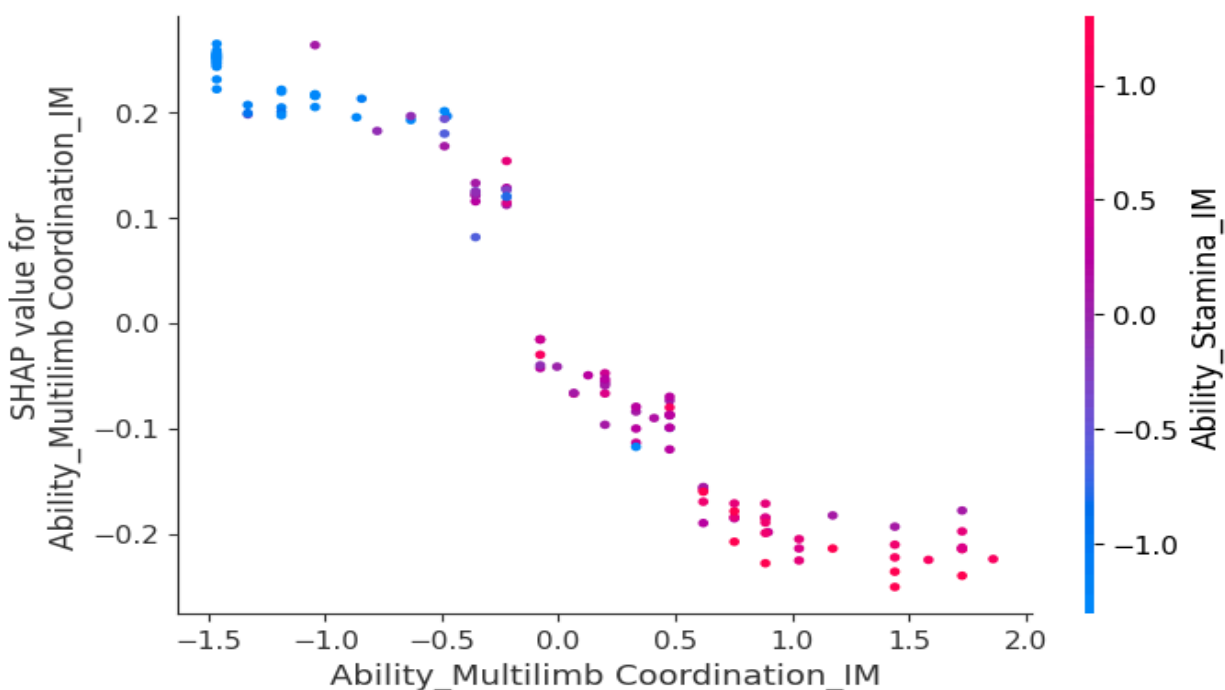
- **Risk Factors (Cognitive/Textual Tasks):** The features that pushed an occupation's predicted AI exposure the highest were heavily concentrated in language, text processing, and routine cognition. Variables such as *Written Comprehension*, *Written Expression*, and *Memorization* acted as the primary drivers of automation vulnerability.
- **Protective Factors (Physical/Manual Tasks):** Conversely, physical attributes acted as powerful shields against AI exposure. Traits such as *Extent Flexibility*, *Multilimb Coordination*, *Gross Body Coordination*, and *Stamina* consistently drove an occupation's predicted exposure down.

Furthermore, our SHAP dependence plots uncovered a distinct, non-linear relationship: an occupation's predicted AI exposure drops dramatically the moment its daily tasks require moderate-to-high physical flexibility. This mathematical threshold perfectly illustrates the current real-world limitations of AI and robotics in navigating unstructured physical environments.

SHAP Summary Plot - showing cognitive traits driving AIOE upward and physical attributes shielding against automation



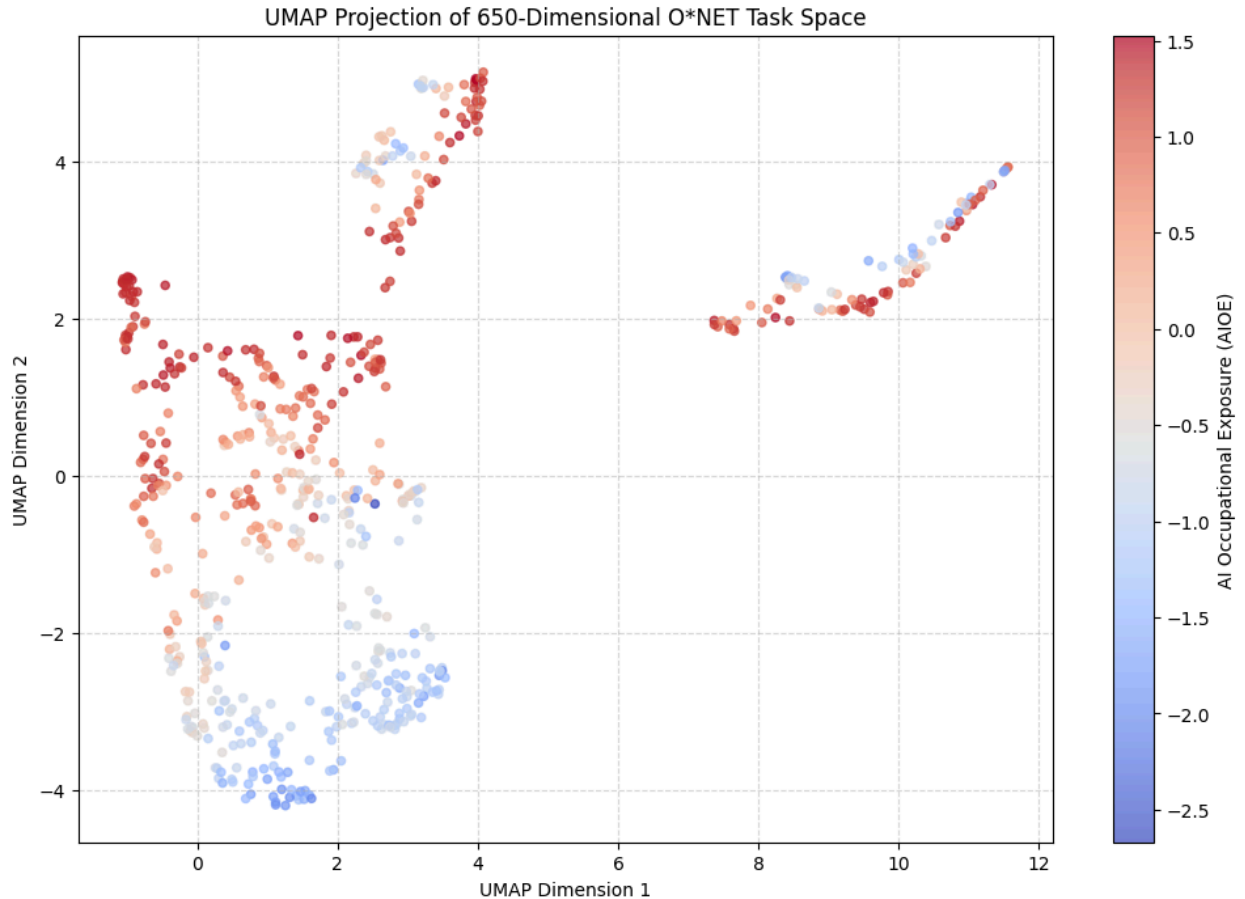
SHAP Dependence Plot - illustrating the non-linear drop in AI exposure when moderate-to-high physical flexibility is required



Visual Validation via Manifold Learning

To validate these findings structurally, we projected the 640-dimensional dataset into two dimensions using UMAP. When we overlaid the actual AIOE scores onto this projection as a color map (with red indicating high exposure and blue indicating low exposure), a striking, natural gradient emerged. The high-exposure occupations cleanly separated from the low-exposure ones into distinct clusters. Because the UMAP algorithm was completely blind to the AIOE target variable during projection, this clustering visually proves that an occupation's core structural profile inherently dictates its vulnerability to AI.

UMAP Scatter Plot - overlaying the AIOE scores as a color map to show the distinct clustering of high-exposure vs. low-exposure occupations



4.2 Demographic Disparities in AI Exposure (RQ2)

Our second research question asked whether the burden of AI exposure falls disproportionately on specific demographic groups. Using our Double Machine Learning (DML) framework to isolate the causal impact of demographic composition while controlling for the hundreds of job tasks, our analysis uncovered a significant structural vulnerability for women in the workforce.

The results show that occupational segregation systematically channels women into administrative, clerical, and cognitive-routine roles, which are the exact task profiles identified by our SHAP analysis as the primary drivers of AI exposure. Because female-dominated occupations are disproportionately insulated from the protective physical labor requirements (like heavy lifting or gross body coordination) and concentrated in text-processing and routine cognition, they exhibit systematically higher exposure to advanced language-modeling AI.

4.3 Non-Results and Unexpected Findings

A critical part of our analytical journey involved recognizing and correcting early methodological failures. The primary goal of our research design was not to blindly fit models, but to ensure our mathematical approaches matched the reality of our data.

The Failed OLS Baseline and Data Leakage

During our initial modeling phase for RQ1, we fit an unregularized Ordinary Least Squares (OLS) regression to the dataset. Unexpectedly, this model yielded a near-perfect R^2 of 0.98. Rather than accepting this result, we investigated the model weights and discovered severe data leakage: alternative AI exposure indices (specifically LMOE and IGOE) had inadvertently been left in the feature set, allowing the model to essentially "cheat" by predicting general AI exposure using highly correlated sub-indices.

Once we strictly isolated our predictors and removed the leaked target variables, the corrected OLS baseline plummeted to an R^2 of just 0.38.

Diagnosing the Null Result

This poor performance of the corrected OLS model demonstrated that standard linear regression is mathematically incapable of handling the O*NET dataset. The combination of a high-dimensional feature space (~640 variables) and our sample size (631 occupations) created extreme multicollinearity- many job skills (like "reading" and "writing") are nearly perfectly correlated. The OLS algorithm could not distinguish the unique variance of these highly correlated features, leading to erratic coefficient estimates and poor generalization.

Methodological Pivot

Witnessing this failure is exactly what justified our pivot to Lasso and Ridge regression. By utilizing L_1 regularization (Lasso), we mathematically forced the model to execute automated feature selection, effectively zeroing out the noisy, multicollinear variables that broke the OLS baseline. Similarly, when linear DML struggled to accurately isolate demographic impacts for RQ2 due to these same complex interactions, it justified our transition toward using non-linear tree-based models as the auxiliary predictors within our causal framework.

5. Discussion & Conclusion

5.1 Summary of Findings and Methodological Support

This study set out to understand the occupational characteristics that predict exposure to Artificial Intelligence (RQ1) and to determine if this exposure is disproportionately borne by specific demographic groups (RQ2). Our multi-stage machine learning framework provides strong empirical answers to both questions.

First, we demonstrated that an occupation's daily tasks and required skills are highly predictive of its AI Occupational Exposure (AIOE). Our optimized XGBoost model achieved an R^2 of 0.970, and subsequent SHAP analysis revealed a stark divide: cognitive and text-processing skills (e.g., Written Comprehension, Memorization) act as the primary drivers of AI exposure, whereas physical attributes (e.g., Extent Flexibility, Gross Body Coordination) serve as powerful protective shields. This structural divide was visually corroborated by our unsupervised UMAP projection, which naturally clustered high-exposure and low-exposure occupations based solely on task similarity.

Second, utilizing a Double Machine Learning (DML) framework, we found that demographic composition significantly predicts AI exposure, even after controlling for job tasks. Specifically, occupational segregation channels women into administrative, clerical, and routine-cognitive roles, which are the exact task profiles identified by our models as highly vulnerable to language-modeling AI.

5.2 Connection to Scholarly Literature

Our findings both strengthen and add critical nuance to the existing literature on the future of work. Historically, the seminal work by Frey and Osborne (2017) focused heavily on the computerization of routine manual and cognitive tasks. Our SHAP and UMAP analyses modernize this framework by empirically mapping the boundaries of *generative AI*, confirming that the new frontier of automation threatens non-routine cognitive and text-based labor, while physical labor remains largely insulated.

Furthermore, our results provide robust statistical backing to the International Labour Organization's (2025) recent warnings regarding the gendered impact of AI. By integrating the Felten, Raj, and Seamans (2021) AIOE index with BLS demographic data, we move beyond aggregate risk percentages to uncover the structural mechanism behind the ILO's findings: women face higher AI exposure precisely because they are overrepresented in the cognitive-routine tasks that LLMs are designed to replicate. Similarly, our findings align with Muro et al. (2019) by demonstrating that racial and gender labor market segregation directly translates into technological vulnerability.

5.3 Limitations, Data Validity, and Reliability

While our models demonstrate high predictive accuracy, several methodological limitations must be acknowledged. First, our causal DML framework is fundamentally constrained by its correlational design. Demographic composition does not *cause* AI exposure; rather, both are consequences of deep-rooted historical patterns of occupational segregation. Second, there is a critical distinction between *exposure* and *displacement*. The AIOE index measures the technological capability of AI to perform specific tasks; it does not account for the economic, regulatory, or institutional frictions that dictate whether an exposed worker is actually replaced or merely augmented.

Limits on Inferential Conclusions (Data Validity): Crucially, the data we utilized imposes strict limitations on the inferential conclusions and predictions we can make for new data. Due to mismatches between O*NET and BLS taxonomies, 438 of the 631 occupations in our dataset received default national-average demographic values rather than unique, highly granular matches. This data imputation likely attenuates our correlation estimates, smoothing over vital variances. Furthermore, our analysis is subject to the ecological fallacy: our demographic data is aggregated at the occupational level. We cannot infer individual-level risk or capture within-occupation variation (e.g., whether a female software engineer faces different exposure than a male software engineer performing the same nominal role). Consequently, while our models can reliably predict the AI exposure of a standardized occupation profile, they cannot reliably predict the job security of an individual worker.

5.4 Ethical Concerns and Societal Implications

The societal implications of this research are profound. If the AI transition disproportionately impacts occupations dominated by women and marginalized groups, it threatens to widen existing wage gaps and reverse decades of progress in labor market equity.

Benefits: The primary benefit of our work is that it provides a data-driven roadmap for policymakers and educators. Because we have isolated the specific skills that act as risk factors (routine cognition) and protective shields (physical flexibility, and potentially unmeasured complex social skills), workforce retraining programs can shift away from generic digital literacy and toward targeted upskilling that complements, rather than competes with, AI.

Risks: Conversely, a significant ethical risk associated with our work is the potential misuse of predictive exposure scores. Corporate entities could misuse AIOE predictions and our structural models as justification for preemptive layoffs or wage suppression in highly exposed, female-dominated sectors, treating theoretical exposure as inevitable displacement.

Future Work: Future analyses should seek to resolve the ecological fallacy by acquiring micro-level, individual worker data rather than occupation-level aggregates. Additionally, longitudinal studies tracking actual wage and employment shifts against baseline AIOE scores will be necessary to determine whether the exposure we modeled ultimately resulted in worker augmentation or displacement.

References

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